

Measuring Cross-Lingual Robustness for Chemistry-Patent Retrieval: Patent-Grounded Multilingual Benchmarks, New Metrics, and a Deployment Decision

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Abstract

A chemistry-patent search team that must deploy a single multilingual embedding model typically ranks candidates by average Recall@10—the one number that hides the failure they will ship. Average recall is inflated by *same-language* hits; the moment a query must reach a patent in another language (the normal case in a patent family), recall collapses, and no two language versions of the same question return the same documents. We make this collapse measurable. We release two *content-controlled*, patent-grounded multilingual chemistry-retrieval benchmarks built only from human-translated patent text—an alias-graph benchmark (132 questions, 24 ChEBI compounds, with chemically-confusable hard negatives) and a cross-lingual (CLIR) benchmark (137 questions over a 23,487-document shared corpus, with Spanish as a pure “no-home” query language)—and a robustness-metric family that reports cross-lingual recall, ranking consistency, confusion, and embedding separability *next to* recall. Across nine models, best cross-lingual Recall@10 is only 0.50, the same-language home advantage reaches +0.55, five language versions of a query agree on barely a third of their top-10 (cross-lingual RBO ceiling 0.39 on the alias-graph benchmark, 0.19 on the cross-lingual benchmark), and a wrong but confusable compound out-ranks the right one on 14–78% of queries. We further quantify *what cross-linguality costs*: to find a foreign twin at the median you scan $3.5\times$ as many documents as for a same-language copy (*cross-lingual reading-cost multiplier*, XRC50), a top-100 re-ranker can recover at most 74% of foreign twins (*re-ranker recoverability ceiling*, RRC@100), and 5.8% are structurally unrecoverable by any re-ranker (L_∞)—a measured share recoverable only by alignment, not re-ranking. We show the cause is an embedding-level separability deficit ($r(\text{cross-language AUC, CLIR@10}) =$

+0.96, robust to dropping the two collapsed encoders), so the lever is representation alignment at indexing time, not a monolingual re-ranker at query time. On a cost-vs-capability frontier, embeddinggemma is the Pareto-optimal capability corner (not the cheapest reader), and machine-translating the *question* is safe (no significant penalty), so human-translation budget belongs on the corpus.

1 Introduction

A chemistry-patent search team that goes multilingual faces a deceptively simple engineering decision: pick *one* dense multilingual embedding model to index a cross-jurisdictional patent corpus, and ship it. The dashboard that guides that choice almost always shows a single number—mean Recall@10 on a held-out query set—and the team picks the model on top. We argue that this number is the wrong number, and that it systematically hides the failure the team will actually deploy.

The reason is structural. In a multilingual patent corpus, the same invention appears in several languages, and a query in one language must routinely reach a document in another (a German chemist searching an English or Chinese family member). Average recall, however, is dominated by the *easy* same-language case: when a same-language copy of the answer exists, every model finds it. The hard, deployment-relevant case—retrieving a *foreign* language version—is averaged away. Our cross-lingual benchmark makes this concrete by construction: it contains 34 Spanish queries with *zero* Spanish gold documents, a built-in “no-home” stress test that average recall cannot see.

Two distinct failures hide behind that average, and both are deployment bugs. First, **inconsistency**: ask the same compound in five languages and the five ranked lists barely overlap—the best cross-lingual rank-biased-overlap (RBO)

any of our nine models achieves is only 0.39 on the alias-graph benchmark and 0.19 on the cross-lingual benchmark, meaning two language variants of a query share fewer than four of their top-10 documents. Second, **confusion**: a chemically-confusable *wrong* compound out-ranks every correct patent on 14–78% of queries depending on the model. A search team that reports only mean recall sees neither. And both are *costly*: at the median, finding a foreign twin takes scanning $\sim 3.5\times$ more documents than a same-language copy (§6), the deficit lives at the embedding level—so the fix is alignment, not re-ranking—and the model choice is a cost-vs-capability frontier, not a single cheapest model.

We build the instruments to measure both failures, quantify what they cost, identify the model that survives them, and trace the cause to the embedding level. Our contributions are:

- **C1.** The first *content-controlled*, chemistry-ontology-grounded cross-lingual patent-retrieval benchmarks, built only from human-translated patent text: an *alias-graph* benchmark whose gold is defined by a chemical ontology and whose hard negatives are confusable neighbours, and a *cross-lingual* benchmark over a 23,487-document shared corpus whose gold is genuinely parallel human-translated patents—not publication_number equivalence and not machine-translated documents (§3).
- **C2.** A cross-lingual robustness-metric family reported co-equally with recall—CLIR@k and the home advantage gap, the directional CLIR matrix (CLIRMatrix-style), mate-retrieval, cross-lingual RBO, language-collapse, and embedding separability AUC—anchored by two new deployment-cost instruments: *XRC*, the cross-lingual reading-cost multiplier, presented on a capability-conditioned (*XRC*50, CLIR@10) Pareto frontier, and *RRC*, the re-ranker recoverability ceiling, read as a per-model budget curve with a knee K^* and a structural floor L_∞ , which an *alignment-recoverability index* (ARI) decomposes into a re-ranker-recoverable, a deeper-pool, and an alignment-only share that sum to one. A reproducible degeneracy gate (CLIR@10 < 0.10) defines which models are excluded; we use the CLIR-

MRS / MRS composite only to order table rows, not as a contribution (§4).

- **C3.** A mechanism finding, confirmed on a content-controlled corpus and made falsifiable: availability sets the stage but a residual encoder bias remains, the modal confusion is a same-language sibling, and the collapse is an embedding-level separability deficit—so a monolingual re-ranker cannot recover underscored foreign twins, a claim we bound per-model with RRC (§7).
- **C4.** A concrete deployment recommendation with an operating rule, grounded in a capability-conditioned cost frontier and a re-ranker-budget knee rather than mean recall or a hand-weighted composite: deploy embeddinggemma (the Pareto-optimal capability corner, with bge-m3 the cheaper-to-read alternative), report XRC/RRC/CLIR@10/language-parity next to recall, budget the re-ranker by the knee, do not reflexively ensemble, and spend human-translation budget on the corpus rather than the queries; a per-route frontier shows the single best model is route-dependent, so a per-language router is genuine upside headroom rather than a delivered win (§8).
- **C5.** A reproducible QAC generation and audit pipeline the benchmarks rest on, with human validation summarized in the system description (§3, Appendix).

In one line: report robustness, not mean recall; and fix cross-lingual retrieval by aligning representations at index time, not by re-ranking at query time. Figure 1 previews the gap that motivates the paper—every model loses ground when no same-language document exists.

2 Related Work

Multilingual and cross-lingual leaderboards—MIRACL (Zhang et al., 2023), MMTEB (Enevoldsen et al., 2025a), the TREC NeuCLIR track (Lawrie et al., 2023), MTEB (Muennighoff et al., 2023), and the directional CLIRMatrix (Sun and Duh, 2020) whose decomposition style we apply to parallel patents—report a *single averaged* score per model, are themselves aggregation-sensitive (Gjorgjevikj et al., 2026; Enevoldsen et al., 2025a), and motivate fresh content-controlled

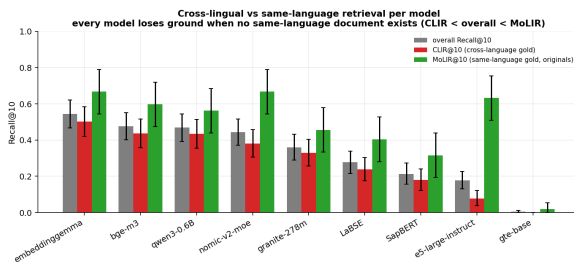


Figure 1: Cross-lingual vs. same-language retrieval, nine models. Overall Recall@10 (grey) sits between the easy same-language case (MoLIR@10, green) and the hard cross-lingual case (CLIR@10, red). The best cross-lingual Recall@10 is only 0.50 (embeddinggemma); average recall hides the collapse. Source: chem-patents key findings, Fig. 1.

assets (Enevoldsen et al., 2025b); ours is instead a ranking-level robustness suite reported coequally with recall, tested across four aggregation schemes (§6). Cross-lingual RAG and language-preference studies—BordIRlines (Li et al., 2024), XRAG (Liu et al., 2025), Linguistic Nepotism (Ki et al., 2025)—confound *content* with *language* (non-parallel Wikipedia or machine-translated imports), whereas our parallel *human-translated* patents isolate the language effect (the C1 defense). Two studies reach a close thesis on general benchmarks—*What Drives Cross-Lingual Ranking?* (Goworek et al., 2025) attributes the gap to weak alignment over translation, and *The Cross-Lingual Cost* (Amiraz et al., 2025) (Arabic–English) documents the same-language head start—which we *confirm*, not discover, on a content-controlled patent corpus that removes their confounds. Conformal/ calibration-IR tools (Li et al., 2023; Rouzrokh et al., 2024; Feng et al., 2025) and fairness post-processing (Xian et al., 2023; Buyl and Bie, 2022) are complementary but score calibration, not ranking robustness. For patent IR, the PRIME NTCIR line (Fujii and Ishikawa, 2002) and the canonical multilingual CLEF-IP (Piroi et al., 2011) use prior-art gold and are not chemistry-grounded, while DAPFAM (Ayaou et al., 2025) uses the family-level gold we reject because family members “may differ in their claims” (so publication_number-equivalence is unsafe); we instead define gold by ChEBI (Hastings et al., 2016) membership and by genuine parallel human translation. Chemistry/entity encoders SapBERT (Liu et al., 2021) and PaECTER (Ghosh et al., 2024) are monolingual, and a concurrent study (Yousefiramandi and

Cooney, 2026) ranks the same model family on *English-only* patents (a different ordering). Finally, whether to translate query or documents is a 25-year CLIR question (Oard, 1998; Saleh and Pecina, 2020; Kishida, 2008) that our budget rule (§8) re-derives for embedding retrieval, and we apply retrieval-consistency analysis (Bailey et al., 2017) cross-lingually as the basis for cross-lingual RBO.

3 Benchmarks

We release two benchmarks, both grounded in real chemistry-patent text and both deliberately avoiding the two shortcuts that make a “multilingual” benchmark misleading: machine-translated source documents (translationese) and family-number gold (non-equivalence).

Shared corpus. Both benchmarks retrieve against one shared haystack, the multilingual_GP corpus of **23,487** documents drawn from Google Patents and EPO bulk data in English, German, Spanish, French, and Chinese, so that results are directly comparable across benchmarks.¹ Detailed corpus-construction statistics (source counts after de-duplication; IPC/A61 distribution) are deferred to the system description; all load-bearing sizes below come from the two benchmark datasets.

Alias-graph benchmark. Here a query is a multilingual ChEBI alias set for a compound (e.g. tin: *tin*/Sn, *Zinn*, *estaño*, *étain*), the gold set is the patents about that concept, and the hard negatives are documents about chemically-confusable neighbours (siblings, parents, tautomers) taken from the compound’s alias graph. The benchmark has **132** questions over **24** ChEBI compounds in five languages, and reports *two co-equal relevance lenses*: a *concept* lens (any patent about the compound; ~109 gold per query) and a *per-publication* lens (the query’s own source patent and its translations; ~2.4 gold per query).² The per-publication lens is the one we read as a quality score; the concept lens is capped by its ~109 positives and must not be read as recall quality.

Cross-lingual (CLIR) benchmark. Here a query is a generated question about a source patent, and the gold set is *every language version*

¹Corpus row count, chem-patents key findings header.

²Alias benchmark size and lens definitions, alias-graph key findings header.

of that source patent. The benchmark has 137 questions in English, German, Spanish, French, and Chinese, split into two non-symmetric populations: 57 **original** questions, asked in the source patent’s own language (so each has exactly one same-language gold), and 80 **synthetic** questions, machine-translated into another language (so they have *no same-language gold at all*—pure cross-lingual retrieval).³ **Spanish is a pure query-side language:** 34 Spanish queries map to zero Spanish gold documents, giving us a controlled “no-home” setting in which average recall has nowhere easy to hide.⁴

Honesty by design. Two choices keep the benchmarks honest. (i) *Source text is human-translated only.* We never use machine-translated documents in the corpus, so a cross-lingual miss is a retrieval failure, not a translationese artifact. (ii) *Gold is defined by ontology or by human translation, never by publication_number,* side-stepping the patent-family non-equivalence problem (§2). Machine translation is used only to translate *questions* for the synthetic population, and §6 shows that this introduces no significant cross-lingual penalty.

Generation and validation pipeline (C5). The queries are produced by a reproducible per-(document, language, mode) generation pipeline that emits three scored candidates best-first, graded on faithfulness and answerability, across technical and semantic question modes. A human annotator reviewed a sample of the generated items; the human-rated quality and the LLM auto-grader’s strictness relative to humans are summarized qualitatively in the system description. We keep C5 as supporting evidence: the benchmarks’ validity rests on the design choices above, not on these validation numbers.

4 Metrics

Recall measures *capability*; it does not measure *robustness to language*. We define a family of metrics that isolate the cross-lingual axes a single recall number averages away. All are computed per model on both benchmarks.

³Population split and Spanish property, chem-patents key findings header and “defining property.”

⁴Benchmark defining property, chem-patents key findings.

CLIR@k, MoLIR@k, and home advantage. We split Recall@k by whether the retrieved gold is in the query’s own language: *MoLIR@k* restricts to same-language gold, *CLIR@k* to cross-language gold, and the *home advantage* MoLIR–CLIR is the same-language head start—the direct instrument for “average recall hides collapse.”

Directional CLIR and asymmetry. A query-language $\rightarrow$ document-language recall matrix and its asymmetry (CLIRMatrix-style (Sun and Duh, 2020)), since reaching language b from a query in a need not equal the reverse; it shows which targets are easy and which directed pairs are most asymmetric.

Mate-retrieval. For a query whose answer also exists as a foreign *twin* (a parallel translation), *mate-hit@k*, *mate-MRR*, and the first-foreign-twin rank measure whether the twin is found at all and at what depth—separating “findable” from “ranked first.”

Cross-lingual RBO. Rank-biased overlap (Webber et al., 2010) between the five top- k lists of a compound asked in five languages (RBO= 1 identical, 0 disjoint), treating the languages as query variants—our *consistency* axis: does the index return the same patents regardless of query language.

Language collapse / over-representation. How much a query language over-fetches its own language relative to the corpus base rate, and how often same-language *noise* out-ranks foreign *gold*—the bias mechanism behind the home advantage.

Separability AUC. $\text{AUC}(\text{score}(\text{gold}) > \text{score}(\text{non-gold}))$ computed separately for same- and cross-language gold; a low cross-language AUC means foreign gold is *under-scored*, not merely mis-ordered—the distinction that decides whether a re-ranker can help.

XRC: cross-lingual reading-cost multiplier. With D_C^{same} and D_C^{cross} the retrieval depths at which a coverage fraction C of same- and cross-language gold (foreign twins) is reached,

$$\text{XRC}_C(m) = D_C^{\text{cross}}(m) / D_C^{\text{same}}(m) \quad (1)$$

is the factor by which one must read deeper to reach a foreign twin than a same-language

copy—a cost in *documents read* that is monotone-invariant, unlike an AUC or composite. Computed population-level (D^{same} over the 57 same-language-gold queries, D^{cross} over the 137 cross-language-gold queries), we report **XRC50** (the median) as the headline because it is finite at our sample size; D_{90}/D_{95} are right-censored and reported only as lower bounds.

RRC: re-ranker recoverability ceiling. A re-ranker can only reorder the pool it is given, so

$$\text{RRC@}K(m) = \Pr[\text{first foreign twin rank} \leq K] \quad (2)$$

(the cumulative first-foreign-twin hit rate over cross-lingual queries) bounds it: $1 - \text{RRC@}K$ is *provably unrecoverable* by any top- K re-ranker. The diminishing-returns-with-depth shape is the established cascade result (Nogueira et al., 2019; Gao et al., 2021); our contribution is reading it as a per-model budget curve with a knee K^* and a structural floor $L_\infty = 1 - \text{RRC@}K_{\max}$ that no re-ranker over the top-1000 can move—turning “align, don’t re-rank” into a per-model upper bound.

ARI: the alignment-recoverability decomposition. At depth K a model’s cross-lingual shortfall splits into a re-ranker-recoverable part ($\text{RRC@}K$), a deep-pool part ($\text{RRC@}1000 - \text{RRC@}K$), and the alignment-only floor $L_\infty = 1 - \text{RRC@}1000$; the three sum to one. We report the scalar

$$\text{ARI@}K(m) = \frac{L_\infty(m)}{1 - \text{RRC@}K(m)}, \quad (3)$$

the fraction of the *remaining* gap after a top- K re-rank that only alignment can move. Normalizing a re-ranking remainder has a precedent in long-tailed reranking (Wang et al., 2026); we invert it (the un-rerankable share) and tie it to alignment, with an alignment-only floor it has no analogue for. ARI is the quantitative form of “align, don’t re-rank” (Appendix Fig. 9). Source: `extra_ari_decomposition`.

The degeneracy gate. XRC, RRC, and separability are meaningless for a model that retrieves almost nothing, so we exclude such models with one reproducible criterion: a model is *degenerate* iff $\text{CLIR@}10 < 0.10$, which flags *exactly* gte-base and e5-large-instruct (Appendix Fig. 11) and

anchors every later “non-degenerate” qualifier.⁵

CLIR-MRS / MRS (table-ordering convenience only). For completeness we report a composite. On the cross-lingual benchmark,

$$\text{CLIR-MRS} = \text{cap} \times (0.5 + 0.5 \cdot \text{rob}), \quad (4)$$

where capability `cap` aggregates accuracy, CLIR, and separability and robustness `rob` aggregates consistency, MT-robustness, and language-parity; the alias-graph MRS is the min-max-normalised mean of accuracy, consistency, confusion-robustness, language-parity, and separability (95% bootstrap CI). We use these *only to order table rows*—the per-axis numbers carry the argument and §6.3 shows the ranking is aggregation-sensitive, so no composite weighting is load-bearing.

5 Experimental Setup

Models. We evaluate **nine** multilingual embedding models as drop-in dense retrievers: embeddinggemma, bge-m3 (Chen et al., 2024), granite-278m, nomic-v2-moe, qwen3-0.6B, LaBSE (Feng et al., 2022), SapBERT (Liu et al., 2021), e5-large-instruct, and gte-base.

Protocol. Both benchmarks retrieve against the single shared multilingual_GP haystack (23,487 documents); queries and qrels come from the respective benchmark datasets. We report Recall@10 and the full metric family of §4. For the alias-graph benchmark we report both relevance lenses; for the cross-lingual benchmark we report the original/synthetic split.

Reproducibility. Every figure and number regenerates from `run_all.py` and the `experimental_plots/extra*.py` scripts under the two run families (`reports/runs/{chem_patents,alias_graph}/`), which also emit the executive summaries this paper cites; the full script list is in Appendix D.

6 Results

We report the cross-lingual benchmark findings first (§6.1), then the alias-graph findings (§6.2), then the two leaderboards (§6.3).

⁵A stricter two-criterion gate ($\text{CLIR@}10 < 0.10$ and $\text{RRC@}1000 < 0.10$) flags only gte-base; the single CLIR@10 rule recovers both exclusions. Source: `extra_two_tax_degeneracy`.

6.1 The cross-lingual collapse

Collapse is real and large. Cross-lingual recall trails same-language recall in *every* model (Figure 1). The best CLIR@10 is 0.50 (embeddinggemma), and the home advantage MoLIR–CLIR reaches +0.55 for the most biased model (e5-large-instruct)—much of which gold availability shapes (§7), though a residual encoder bias remains: a same-language copy of the answer is far easier to retrieve than its foreign twin (Appendix Fig. 5).⁶

Retrieval is anisotropic: a hub-and-spoke graph. Read as a directed graph (Appendix Fig. 6), the query-language $\rightarrow$ document-language matrix is not symmetric: the hardest directed edge is $\text{en} \rightarrow \text{de}$ ($\text{R@10} = 0.12$) and the most asymmetric pair is $\text{de} \leftrightarrow \text{zh}$ (gap +0.23). Contrary to a common assumption, *no single language is a clean “easiest target”*: pooled over the eight models with a defined cross-lingual recall (all but the degenerate gte-base), French (0.375) and English (0.367) are nearly tied as targets (within 0.01 of each other), with Chinese (0.350) and German (0.309) behind. What asymmetry exists partly tracks corpus composition (en 46% vs. zh 0.4% of documents), not only encoder behaviour—there are simply few German or Chinese documents to retrieve. A single “recall” number erases all of this directionality.

The cost of cross-linguality, and the cost frontier. How much does the collapse cost in practice? At the median, the best model embeddinggemma reaches a same-language gold at depth 2 but its foreign twin only at depth 7—an XRC50 of 3.5 $\times$ (Eq. 1). The cleanest model on this axis is granite-278m (1.25 $\times$); the cost balloons for nomic-v2-moe (11.5 $\times$) and becomes catastrophic for e5-large-instruct (97.75 $\times$), while gte-base is degenerate (§4: CLIR@10 < 0.10) and its XRC is undefined.

A low reading cost is worthless if the model retrieves nothing, so XRC is not a scalar to minimize: we read it *against* capability on the standard cost-vs-capability Pareto frame (Conway et al., 2025), but with XRC—our new reading-depth multiplier—as the cost

axis in place of the usual latency or parameter count, giving (XRC50, CLIR@10) space (Figure 2). Three models are Pareto-optimal—no other model is both cheaper-to-read and higher-CLIR: embeddinggemma (XRC50 3.5 $\times$, CLIR@10 0.50), bge-m3 (2.0 $\times$, 0.44), and granite-278m (1.25 $\times$, 0.33). embeddinggemma is the frontier’s *unique maximum-CLIR corner*; it is *not* the cheapest deployable model. At a stated (untuned) deployment threshold $\tau=0.40$ the admitted set is {bge-m3, qwen3-0.6B, embeddinggemma} and the cheapest-to-read admitted model is bge-m3 (2.0 $\times$ vs. 3.5 $\times$), not embeddinggemma. This snapshot survives only over a narrow band: sweeping τ , the admitted set is stable for $\tau \in [0.385, 0.43]$ and bge-m3 remains the cheapest admitted reader for $\tau \in [0.33, 0.435]$; below $\tau \approx 0.33$ the cheaper-to-read granite-278m (1.25 $\times$) enters the admitted set and the cheapest-reader recommendation *flips* to granite-278m, while above $\tau \approx 0.45$ only embeddinggemma qualifies. Only embeddinggemma’s status as the unique maximum-CLIR@10 corner is τ -invariant. The picture also reads off a cautionary point: the cheapest reader *can* be among the worst retrievers (the dominated look-alikes like SapBERT read shallow precisely because they surface little)—a directional read-off of the figure, not a statistical claim (the XRC50~CLIR@10 rank correlation across the seven non-degenerate models is $\rho=+0.29$, $n=7$, $p=0.53$, n.s.). The deployment consequence is a *frontier choice*, not a single cheapest model (§8).

Machine-translating the question is safe. Controlling for the source patent, the paired human–MT difference in cross-lingual reach is -0.044 ($p = 0.13$): statistically *insignificant* (Appendix Fig. 7). We read this as a *null* result—MT of the query does not significantly hurt and we do not claim it helps—which validates the project’s “MT-is-fine-for-the-question” assumption.

Foreign twins are buried, and a re-ranker cannot reach them all. Pooled mate-hit@10 is 0.38, and on 15% of (query, model) pairs the foreign twin never surfaces even in the top-1000 (Appendix Fig. 8). The best twin-finder, embeddinggemma, reaches its first foreign twin at a median rank of 5 (Appendix Fig. 8)—most models do far worse. This directly bounds any downstream re-ranker (Eq. 2), and we read the bound as a *budget curve* (Figure 3): for embeddinggemma

⁶MoLIR@10 is defined only on the 57 original queries (the only ones with a same-language gold); the home advantage is measured *paired* within those queries, so the +0.55 is a clean within-query contrast even though the green MoLIR bar in Figure 1 is a smaller population than the grey/red bars.

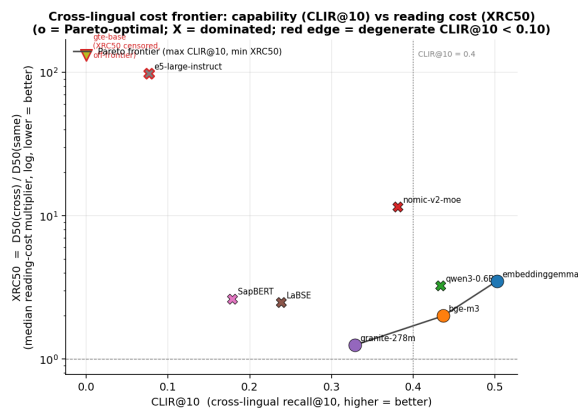


Figure 2: Cost-vs-capability frontier in (XRC50, CLIR@10) space. The Pareto frontier (line) has three members {embeddinggemma, bge-m3, granite-278m}; embeddinggemma is the unique maximum-CLIR corner. Dominated finite models carry an $\times$; gte-base is off-plane (XRC undefined). embeddinggemma is the capability corner, *not* the cheapest reader—at $\tau=0.40$ that is bge-m3 ($2.0\times$), stable for $\tau \in [0.33, 0.435]$ and flipping to granite-278m below $\tau \approx 0.33$; only the embeddinggemma corner is τ -invariant. Source: chem-patents extra_cost_frontier.

the $RRC(K)$ knee is $K^*=5$, so most of the recoverable payoff sits in a shallow top- K and a deeper pool past the knee buys almost nothing; the knee is ≤ 20 for nearly every non-degenerate model. The ceilings are $RRC@100 = 0.7445$ and $RRC@1000 = 0.9416$, so a realistic top-100 re-ranker recovers at most 74% of foreign twins (leaving $\sim 25\%$ on the table) and the floor $L_\infty = 1 - RRC@1000 = 0.058$ (5.84%) of foreign twins is **unrecoverable by any re-ranker over the top-1000**—a floor only representation alignment can move (§7). This floor ranges from 0.058 (embeddinggemma) to 0.372 for the degenerate e5-large-instruct, which the gate excludes; the regression checks behind the curve all pass, so we state it firmly. Re-ranking reads a list retrieval already truncated.

How much of the gap is alignment-bound: the ARI decomposition. The ARI decomposition (Appendix Fig. 9) reports this split per model. For embeddinggemma the alignment-only floor is the smallest of any non-degenerate model ($L_\infty = 0.058$, next qwen3-0.6B at 0.073). On the post-re-rank residual itself embeddinggemma and qwen3-0.6B are *tied* for the lowest re-ranker-irreducible share— $ARI@100 = 0.229$ and 0.233 respectively (Eq. 3), a gap of 0.004 whose 95%

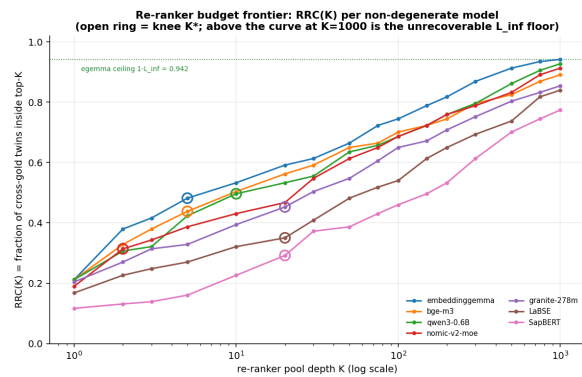


Figure 3: Re-ranker budget curves: $RRC(K)$ vs. candidate depth K (log scale) for the eight models with a defined RRC curve (all but gte-base, whose candidate pool is empty), with the knee K^* (rings) past which a deeper pool buys almost nothing. embeddinggemma knees at $K^*=5$, reaches $RRC@1000 = 0.942$, and leaves an alignment-only floor $L_\infty = 0.058$ unrecoverable by any re-ranker. Source: chem-patents extra_rrc_budget_frontier.

resample interval $[-0.174, 0.176]$ straddles zero (Appendix D). The decomposition is what turns “align, don’t re-rank” from a slogan into a measured per-model remainder.

6.2 Inconsistency and confusion (alias-graph)

If §6.1 measured what cross-linguality costs to *read* (the XRC reading-cost tax), the alias-graph benchmark measures what it costs in *precision*—the second line-item of the same bill: a look-alike compound that out-ranks the gold (a confusability tax). The two taxes are only weakly—and if anything inversely—rank-correlated across the seven non-degenerate models (Spearman $\rho = -0.59$, $n=7$, $p=0.16$, n.s.; Appendix Fig. 20), so neither benchmark is a clean proxy for the other. We read this descriptively, as the motivation for reporting both benchmarks rather than one: a model can read cheaply yet confuse heavily, or vice versa (embeddinggemma is low on both; granite-278m reads cheapest at $1.25\times$ but confuses at 0.182). It is not an independence result.

Same question, different rankings. Asking the same compound in five languages does *not* return the same patents. The best cross-lingual RBO *any* of the nine models reaches is only 0.39 (Appendix Fig. 13)—a ceiling no model beats; two language variants of a query share fewer than four of their top-10 documents, and Chinese is the consistent odd-one-out. The embedding space is far from language-agnostic.

Confusable look-alikes out-rank the gold.

A chemically-confusable wrong compound out-ranks every gold patent on 14–78% of queries (per-publication lens) depending on the model, worst for German and Chinese (Appendix Fig. 12). The damage is concentrated in near-twins: pooled, a *sibling* compound out-ranks the gold 18.1% of the time versus only 6.2% for a more distant *parent* class (a $2.9\times$ ratio; for embeddinggemma, 6.1% vs. 1.5%).⁷ The repeat offenders are a small set of universal attractors—*polypeptide*, *methyl*, *ethene*, *hydroxide*, *dioxygen* (Appendix Fig. 14).

6.3 Leaderboards

Table 1 ranks the nine models by CLIR-MRS on the cross-lingual benchmark and Table 2 by MRS on the alias-graph benchmark. embeddinggemma tops both. Note the order changes from Figure 1: ranking by CLIR-MRS rather than raw recall reshuffles the middle of the field (granite-278m and qwen3-0.6B swap), which is the point of the paper—robustness reorders the recall ranking (embeddinggemma leads consistency and separability, not raw recall alone; the per-axis detail is in Tables 1–2 and Appendix Fig. 19).

The ranking is aggregation-sensitive. We caution against over-reading the composite. Across four aggregation schemes (CLIR-MRS, winner-take-all, Borda, and equal-weight), embeddinggemma’s rank *ranges* from 1 to 4: it is rank-1 only when capability is up-weighted, because it leads the three capability axes (accuracy, CLIR, separability) but trails on the robustness axes (Appendix Fig. 19).⁸ Rankings are therefore aggregation-sensitive; our recommendation (§8) rests on *per-axis dominance*, not on any single composite weighting.

7 Analysis

Why does cross-lingual retrieval collapse here, and what does the failure tell us about the fix?

The modal failure is a same-language sibling. The two traps—language bias and chemical confusability—compound. Across the 257

⁷This is a two-level sibling/parent split; the on-disk relation field is binary, so a graded ChEBI hop-distance decay law is left to future work. Source: extra_confusion_severity.

⁸The winner-take-all (axes-won) column is contaminated by a degenerate model: gte-base “wins” MT-robustness and language-parity only because it retrieves almost nothing (CLIR@10 = 0.000, below the degeneracy gate of §4), so its scores are trivially uniform.

model	R@10	CLIR@10	home	sep	CLIR-MRS
embeddinggemma	0.544	0.502	+0.22	0.90	0.71
bge-m3	0.476	0.437	+0.23	0.87	0.70
granite-278m	0.359	0.329	+0.17	0.86	0.67
nomic-v2-moe	0.443	0.381	+0.34	0.86	0.67
qwen3-0.6B	0.468	0.433	+0.19	0.86	0.58
LaBSE	0.277	0.238	+0.21	0.81	0.40
SapBERT	0.212	0.179	+0.18	0.79	0.40
e5-large-instruct	0.178	0.077	+0.55	0.71	0.26
gte-base	0.004	0.000	+0.02	0.49	0.00

Table 1: Cross-lingual benchmark leaderboard. “home” is the home advantage MoLIR–CLIR; “sep” is separability AUC. embeddinggemma wins (CLIR-MRS 0.71, CI [0.67, 0.77]). Source: chem-patents key findings, Verdict table / headline_numbers.csv.

model	pubR@10	RBO	conf	MRS
embeddinggemma	0.67	0.387	0.144	0.991
qwen3-0.6B	0.641	0.349	0.227	0.902
bge-m3	0.604	0.288	0.237	0.813
LaBSE	0.511	0.314	0.295	0.779
granite-278m	0.529	0.289	0.318	0.757
nomic-v2-moe	0.580	0.239	0.215	0.727
SapBERT	0.352	0.142	0.424	0.494
e5-large-instruct	0.237	0.021	0.481	0.239
gte-base	0.047	0.003	0.782	0.000

Table 2: Alias-graph benchmark leaderboard. “pubR@10” is per-publication Recall@10; “conf” is publication-lens confusion rate. embeddinggemma wins (MRS 0.991, CI [0.86, 1.00]). Source: alias-graph key findings, headline table / headline_numbers.csv.

confused concept-lens cases, the single most common failure is a *same-language sibling* out-ranking the gold ($114/257 = 44.4\%$); siblings account for 79.4% of all confusions and same-language winners for 55.6% (Appendix Fig. 15). The encoder is simultaneously biased toward the query’s own language and toward the nearest wrong compound. This joint mode is the spine of the rest of the section, which we now read benchmark by benchmark.

Availability sets the stage; a residual encoder bias remains. On the *alias-graph* benchmark, each language retrieves its own-language gold far better (0.63–0.82) than foreign-language gold (0.35–0.47), and because English and French source patents dominate, 42% of an English query’s gold is reachable in-English versus only 8–10% for German, Spanish, and Chinese.⁹ Availability therefore shapes *which* language’s gold is reachable—but it does *not* explain away the same-

⁹Availability shares and own-vs-foreign recall, alias-graph key findings “mechanism.”

language bias. Regressing each language’s home advantage on its gold-availability share gives a *negative* slope (-0.57 , $n=5$ languages, descriptive): the language with the *least* reachable gold (Chinese, 8%) carries the *largest* home advantage ($+0.47$), so the mean $+0.32$ home advantage is a residual encoder bias, not an availability artifact (Appendix Fig. 16). On the *cross-lingual* benchmark the same bias shows as language collapse: low-resource query languages over-fetch their own language up to $48.7\times$ the corpus base rate, and same-language *noise* out-ranks the gold on 60% of queries (Appendix Fig. 10).

Structure-style questions are the trap. *How* a query is phrased decides whether the model falls into a confuser. *Structure*-style questions are the trap (Recall@10 0.26, confusion 51%) because structure descriptions are exactly what sibling compounds share, whereas *role* questions are far safer (Recall@10 0.60, confusion 25%); a language-independent *formula token* (e.g. H_2S , CO_2) measurably helps ($p < 0.01$), even in Chinese (Appendix Fig. 17). The trap is also where *no* encoder helps: of the 16/132 (12%) queries missed by all nine models, 14 are structure questions about sibling compounds (predominantly French and Chinese)—the universal-blind core needs chemistry-aware help, not more encoders (§8).

Bias and inconsistency are related, but the robust signal is one. The more a model leans on its home language, the less consistent it is across languages— $r(\text{home advantage, RBO}) = -0.85$ (cross-lingual) and $r(\text{same-language share, RBO}) = -0.87$ (alias-graph), two related bias proxies. We report these as *descriptive* observations on $n=9$ models: both relationships are driven by the two degenerate encoders and do *not* survive dropping them (the over-representation $\sim$ CLIR correlation flips sign and the home-advantage $\sim$ RBO correlation collapses on $n=7$).¹⁰ We therefore do not treat them as mechanisms; the one correlation that *is* robust is the separability link below.

It is a separability deficit, so re-ranking cannot fix it. Behind the confusion, the cosine score barely separates gold from sibling: $\text{AUC}(\text{gold} > \text{look-alike}) = 0.55$ for

¹⁰Drop-the-collapsers check, chem-patents extra_correlation_robustness.

confused queries versus 0.70 otherwise (Appendix Fig. 18). On the cross-lingual benchmark the same story holds across models, and this is the section’s load-bearing mechanism: $r(\text{cross-language AUC, CLIR@10}) = +0.96$ (Figure 4), and unlike the two bias correlations its *sign* is robust. Across model-level bootstrap resamples the separability–floor correlation is positive in 99.97% of draws (point $r = 0.96$); with only seven non-degenerate models the confidence interval is correspondingly wide ($[0.73, 1.00]$), so we report the sign as the robust finding rather than its magnitude (Appendix D). “Robust” here means *sign-robust under resampling*, not tightly estimated. Foreign gold is *under-scored*, not merely mis-ordered. This is the crux: a monolingual re-ranker reads a ranked list it cannot repair, because the right foreign document was never scored high enough to appear—and RRC bounds exactly how much it could repair. The budget curve (§6.1) already flattens by the knee $K^*=5$, a top-100 re-ranker recovers at most $\text{RRC@100} = 0.74$, and the floor $L_\infty = 0.058$ of foreign twins is unrecoverable by *any* re-ranker. The ARI decomposition (§6.1) shows this floor is the *only* alignment-bound part: for the deployed model, after a cheap top-100 re-rank, just $\text{ARI@100} = 0.229$ of the remaining gap requires alignment (tied with qwen3-0.6B at 0.233 for the lowest re-ranker-irreducible residual), while embeddinggemma retains the smallest alignment-only floor ($L_\infty = 0.058$). The lever is at the embedding level. We frame this descriptively rather than causally: cross-language AUC and overall Recall@10 are strongly collinear across our seven models, and once Recall@10 is partialled out the separability–CLIR association is no longer significant (partial $r = +0.29$, $p = 0.57$, n.s.), so separability is a descriptive correlate of the cross-lingual floor, not an effect net of general retrieval capability (Appendix D).

8 Deployment Recommendation

The metrics yield a single, actionable decision for a search team.

Deploy embeddinggemma: the capability corner of the cost frontier. We recommend embeddinggemma because it leads the capability axes *individually*—it tops cross-lingual Recall@10 (0.50) and separability and is the best twin-finder (median first-foreign rank 5)—and because it sits on the cost-vs-capability frontier as

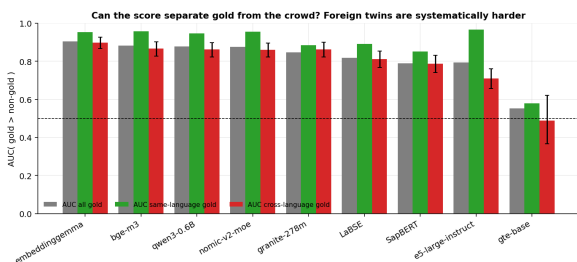


Figure 4: Per model, cross-language gold (red) is systematically harder to separate from non-gold than same-language gold (green); the model-level cross-language-AUC-CLIR@10 correlation is $+0.96$ ($+0.958$ on the $n=7$ non-degenerate models). The *sign* is robust under resampling ($P(r > 0) = 0.9997$); at $n=7$ the 95% CI is wide ($[0.73, 1.00]$), so we report the sign, not the magnitude (Appendix D). Source: chem-patents key findings, Fig. 11; robustness from extra_correlation_robustness.

its *unique maximum-CLIR corner*: no model is both cheaper-to-read *and* higher-CLIR (Figure 2). It is *not* the cheapest deployable model: bge-m3 reads shallower (XRC50 $2.0\times$ vs. $3.5\times$) at a still-admitted CLIR@10, so the decision is a *frontier choice*—pick the capability corner unless the reading budget is binding, in which case bge-m3 is the cheaper-to-read frontier alternative over the stated τ -band ($\tau \in [0.33, 0.435]$; below $\tau \approx 0.33$ the cheaper-reader pick flips to granite-278m, while only the embeddinggemma max-CLIR corner is τ -invariant; §6.1). It also carries the smallest alignment-only floor ($L_\infty = 0.058$) and is tied with qwen3-0.6B for the lowest ARI@100 (0.229 vs. 0.233, gap 0.004 with a 95% CI straddling zero; §6.1), so the part of its cross-lingual gap a re-ranker cannot close is among the smallest of the deployable field. We deliberately do *not* rest the recommendation on the composite: embeddinggemma’s rank ranges $[1, 4]$ across aggregation schemes (§6.3), so a composite-based argument would be fragile. Per-axis dominance and the frontier are not.

The capability corner is route-dependent: a per-route router is upside headroom. Recomputing the $(\text{XRC}_{50\ell}, \text{CLIR}_{10\ell})$ frontier *per query language* ℓ shows the capability corner is not stationary: across the five routes there are *three* distinct max-CLIR corners (en $\rightarrow$ qwen3-0.6B, de/es/zh $\rightarrow$ embeddinggemma, fr $\rightarrow$ nomic-v2-moe), and a recall-only dashboard’s pick *flips* on 2/5 routes (en, fr; Appendix Fig. 21). This nuances, but does *not* overturn, the single-model rec-

ommendation. First, embeddinggemma remains the right default: it is the global capability corner (max pooled CLIR@10 = 0.50), owns the smallest alignment-only floor ($L_\infty = 0.058$) and ties qwen3-0.6B for the lowest ARI@100 (0.229 vs. 0.233, gap CI straddles zero), and per-route wins 3/5 routes (de, es, zh)—including the two hardest cross-only routes (Spanish has *zero* same-language gold; Chinese is the thinnest)—while the recall-only dashboard picks it on all five. Second, a per-route router that swaps in qwen3-0.6B for en and nomic-v2-moe for fr is a plausible *upside*, consistent with the oracle/ensemble headroom below, but on our per-language sample sizes we report it as headroom, not a recommendation. The route corners rest on thin per-language samples (de $n_{\text{same}}=7$, zh $n_{\text{same}}=2$, es $n_{\text{same}}=0$; cross-side $n=22\text{--}34$): the defensible spine is the robust per-route CLIR@10 $_{\ell}$ axis, the frontier membership, and the *existence* of corner movement, whereas the per-route XRC axis is *indicative* only and es is XRC-undefined (zero same-language gold) and never imputed. We therefore present routing as the honest upside of the frontier, not a reversal of the deploy-one-model decision.

Report robustness next to recall. We recommend reporting XRC, RRC, CLIR@10, and language-parity *alongside* recall, never average recall alone. The pure “no-home” Spanish queries are the reason: a dashboard that averages over them will look healthy while the deployed system silently fails on cross-jurisdiction search.

Do not reflexively ensemble. There *is* real headroom—the oracle (any of nine models) reaches CLIR@10 0.61 on the cross-lingual benchmark (Appendix Fig. 22) and 88% versus the best single model’s 76% on the alias-graph benchmark, with Chinese having the largest recoverable headroom (Appendix Fig. 23). But this headroom is *not free*: untuned reciprocal-rank fusion (Cormack et al., 2009) did *not* beat the best single model. The two facts coexist—naive fusion loses, yet a score-aware learned combiner or per-language routing (for the homeless Spanish and the low-resource Chinese) could win. And more encoders will not close the gap on the hardest queries: the 16/132 (12%) universal-blind core is overwhelmingly structure questions about sibling compounds (14/16; §7), which need chemistry-aware help, not another retriever.

Budget the re-ranker by the knee. If a re-ranker is used, the RRC budget curve (Figure 3) sets its depth: most recoverable payoff sits in a shallow top- K —the knee is $K^* \approx 5$ for the deployed embeddinggemma and ≤ 20 for nearly every non-degenerate model—so a deeper candidate pool past the knee buys almost nothing. There is no benefit in paying to re-rank a top-1000 pool when the returns flatten by the top-20.

Align, do not re-rank. Because foreign gold is under-scored at the embedding level (§7), the durable fix is representation alignment at indexing time. A monolingual re-ranker at query time operates on a candidate list that already excludes the right foreign document—and RRC says how much: for the deployed model, $L_\infty = 1 - \text{RRC@1000} = 0.058$ of foreign twins are unrecoverable by *any* re-ranker (the alignment-only floor), and a realistic top-100 re-ranker leaves 25% on the table.

Budget rule. Finally, the MT-of-question null result (-0.044 , $p = 0.13$) tells us where to spend a finite human-translation budget: *machine-translate the query, but human-translate the corpus*. The expensive resource buys the most where it matters—the documents being searched, not the questions. This re-derives the classic query-vs-document translation rule (Oard, 1998; Saleh and Pecina, 2020) for embedding retrieval over patents, where our null quantifies the query-side MT cost as insignificant.

Limitations

Scale and statistical reach. We consolidate the resample interval or sign-stability vote of every load-bearing small- n scalar—the separability r , the three frontier XRC50 medians, the embeddinggemma-vs-qwen3-0.6B ARI@100 tie, and the separability partial- r —in Appendix D, so the honest intervals are reported in one place rather than buried. The benchmarks comprise 132 (alias-graph) and 137 (cross-lingual) questions over 24 compounds; some language-pair cells are thin, so the directional CLIR matrix should be read as indicative rather than definitive in the sparsest cells. For the same reason we report XRC50 (the median) as the robust headline and treat the deeper D_{90}/D_{95} coverages as right-censored lower bounds, not point estimates; and the negative availability slope (-0.57) is a *descriptive*

regression on five language points, not an inferential test. Two further cross-model correlations are reported as descriptive/motivating only, not established effects: the two-tax non-redundancy ($\rho = -0.59$, $n=7$, $p=0.16$) and the cheapest-reader trap ($\rho = +0.29$, $n=7$, $p=0.53$) are both *non-significant* on seven non-degenerate models. The RRC budget objects (knee K^* , floor L_∞) and the ARI decomposition (which closes to 1.0 for all nine models), by contrast, are regression-checked and stated firmly.

Per-route frontier thinness. The per-route capability corners (Appendix Fig. 21) are estimated on thin per-language samples (de $n_{\text{same}}=7$, zh $n_{\text{same}}=2$, es $n_{\text{same}}=0$; cross-side $n=22\text{--}34$). The per-route XRC axis is therefore *indicative* only, and the Spanish route’s XRC is undefined (no same-language gold) and is never imputed. Accordingly we report the per-route router as *headroom*, not a recommendation: the deployed decision is a single model, and the robust per-route signals are the CLIR@10 $_\ell$ axis, the frontier membership, and the existence of corner movement.

Domain transfer. All results are chemistry-patent-specific. We frame that specificity as the contribution—a real deployment niche underserved by general benchmarks—but we do not yet provide a general-domain MMTEB/MIRACL/NeuCLIR companion measurement; it is backlogged.

Composite validation. CLIR-MRS/MRS is a reporting convenience, not externally validated against a downstream utility; the per-axis dominance is what the recommendation stands on. A validation against a held-out human or downstream-RAG criterion is backlogged.

Judge dependence and gold equivalence. The generated queries and the auto-grader rely on an LLM judge. We provide human validation of a sample (Appendix), but claim-level patent equivalence is not lay-annotatable; a larger expert re-annotation, and a spot-check that the parallel human-translated golds are claim-level equivalent, remain future work.

Severity law. Confusion severity is reported as a two-level sibling/parent split; the graded ChEBI hop-distance decay law is future work, since the on-disk relation field is binary.

Language coverage and a forthcoming causal probe. We cover five languages (en/de/es/fr/zh) with a single non-Latin script (Chinese); low-resource scripts are out of scope here. Calibration-style cross-lingual instruments and a conformal (guaranteed-coverage) version of XRC are likewise left as future work; we do not claim them as results. Our “align, not re-rank” conclusion is correlational; the natural next step—which XRC and RRC were built to measure—is to fit a per-language alignment map on one model, re-retrieve, and recompute XRC50/RRC@100/ARI before and after. The ARI decomposition (Appendix Fig. 9) hands this probe a precise before/after target: the per-model L_∞ floor / ARI@100 is exactly the quantity an alignment intervention should move while re-ranking cannot. This is upside only; none of the results above depend on it, and we flag it as forthcoming.

9 Conclusion

For a chemistry-patent search team deploying one multilingual embedding model, average Recall@10 is the wrong dashboard: it is inflated by same-language hits and hides a cross-lingual collapse (best CLIR@10 0.50; home advantage up to +0.55; the best cross-lingual RBO any model reaches is 0.39 on the alias-graph benchmark and 0.19 on the cross-lingual benchmark; a confusable compound beating the gold on 14–78% of queries). That collapse *costs*: finding a foreign twin takes scanning $\sim 3.5\times$ the reading budget (XRC), and a re-ranker cannot fully recover it—the budget curve knees at $K^*=5$ and leaves a floor $L_\infty = 0.058$ that is the only part of the gap a re-ranker cannot move, recoverable by alignment alone (RRC). Our two content-controlled, patent-grounded benchmarks and our CLIR robustness suite make the collapse measurable, place the deployable models on a cost-vs-capability frontier on which embeddinggemma is the Pareto-optimal capability corner (bge-m3 the cheaper-to-read alternative), and trace the cause to an embedding-level separability deficit ($r(\text{cross-language AUC, CLIR@10}) = +0.96$, robust)—so the fix is alignment at index time, not re-ranking at query time, and the budget rule is to machine-translate the query and human-translate the corpus. Reporting robustness next to recall, and treating cross-lingual retrieval as a representation problem, are the two habits we recommend any multilingual patent-search team adopt.

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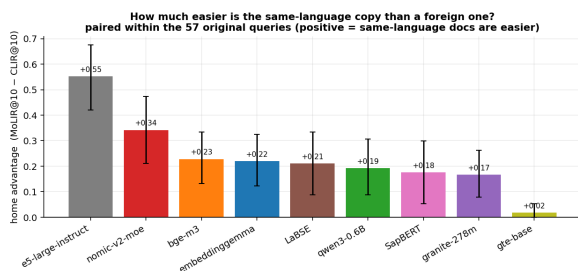


Figure 5: Same-language home advantage (MoLIR–CLIR) per model on the cross-lingual benchmark; it reaches +0.55. Source: chem-patents key findings, Fig. 2.

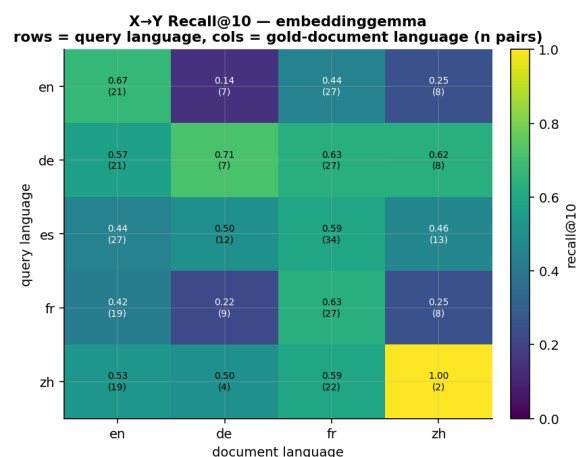


Figure 6: Query→document language recall matrix (best model); retrieval direction matters, but no language is a clean easiest target. The most asymmetric directed pair is de↔zh (gap +0.23; asymmetry panel not shown). Source: chem-patents key findings, Fig. 3; pooled hub scores from extra_directional_hub.

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A Extended Cross-Lingual Results

Figures referenced from §6.1 and §7.

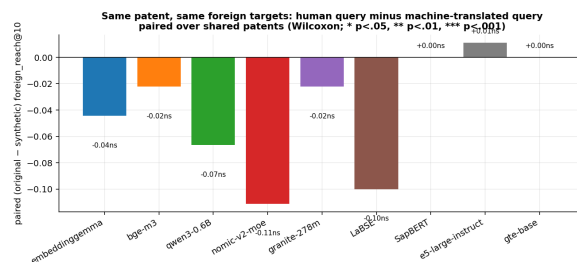


Figure 7: Paired human vs. machine-translated queries; the cross-lingual difference is -0.044 ($p = 0.13$), not significant. Source: chem-patents key findings, Fig. 5.

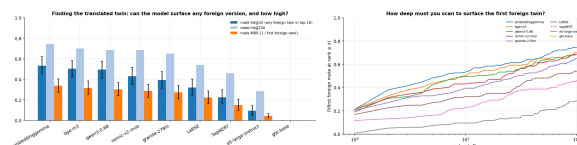


Figure 8: Foreign twins are buried (cross-lingual benchmark). *Left*: mate-retrieval (mate-hit@10/100, mate-MRR); pooled mate-hit@10 is 0.38. *Right*: depth to the first foreign twin; 15% of (query, model) pairs never reach one in the top-1000. Source: chem-patents key findings, Figs. 6–7.

B Extended Alias-Graph Results

Figures referenced from §6.2 and §7.

C Aggregation, Routing, and Ensemble

Figures referenced from §6.3 and §8.

D Robustness Ledger and Reproducibility

Robustness of the load-bearing small- n scalars. Table 3 consolidates the resample interval (or sign-stability vote) of every small- n scalar the body leans on. Three points are deliberate. (i) The separability r is reported by its *sign* ($P(r > 0) = 0.9997$), not its magnitude: at $n=7$ the 95% CI is wide. (ii) The three frontier XRC50 medians have *finite* but wide CIs (censored-draw fraction = 0; median-of-discrete-depth bootstrap), so we report the *depth ordering*, not CI tightness. (iii) The embeddinggemma-vs-qwen3-0.6B ARI@100 gap CI straddles zero (a tie), and the separability–CLIR association is no longer significant once Recall@10 is partialled out—so separability is a descriptive correlate, not an effect net of capability. Source: extra_robustness_appendix/{robustness_table.csv, summary.json}.

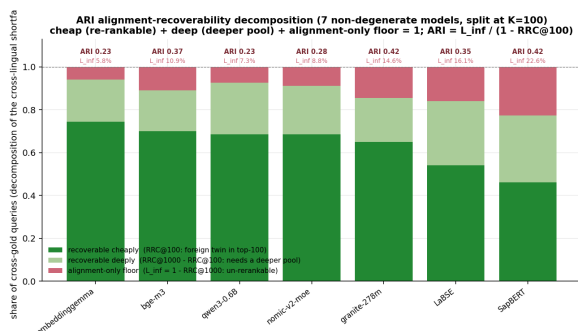


Figure 9: Alignment-recoverability decomposition: each model’s cross-lingual shortfall split into the re-ranker-recoverable part (RRC@100), the deeper-pool part (RRC@1000 – RRC@100), and the alignment-only floor L_{∞} . The three sum to 1.0 for every model (the identity closes for all nine; the figure shows the seven non-degenerate). embeddinggemma has the smallest floor (0.058) and is tied with qwen3-0.6B for the lowest post-re-rank residual (ARI@100 = 0.229 vs. 0.233). Source: chem-patents extra_ari_decomposition.

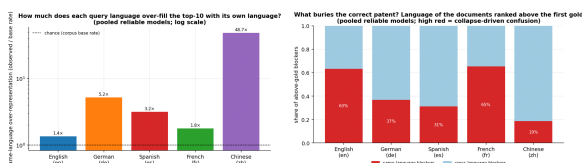


Figure 10: Language collapse (cross-lingual benchmark). *Left*: same-language over-representation by query language, up to $48.7\times$ the corpus base rate. *Right*: the language of the documents that bury the gold; same-language noise out-ranks the gold on 60% of queries. Source: chem-patents key findings, Figs. 9–10.

Reproducibility. The full ten-round derivations behind the headline numbers, the per-language and per-lens tables, and all figures are produced by the scripts under reports/runs/{chem_patents,alias_graph}/experimental_codes/ and the experimental_plots/extra*.py scripts (run_all.py, extra_cost_frontier.py, extra_rrc_budget_frontier.py, extra_two_tax_degeneracy.py, extra_ari_decomposition.py, extra_per_route_frontier.py, extra_directional_hub.py, extra_aggregation_invariance.py, extra_availability_residual.py, extra_joint_failure.py, extra_confusion_severity.py, extra_correlation_robustness.py), summarised in the two

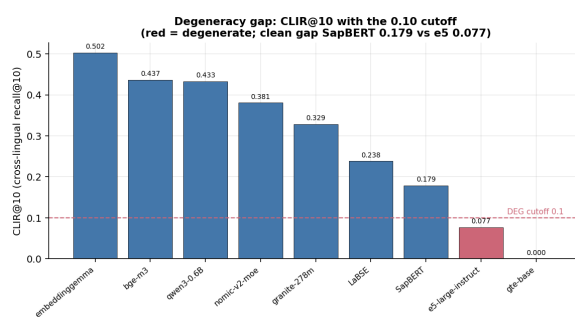


Figure 11: The degeneracy gate $\text{CLIR@10} < 0.10$ (dashed line) separates gte-base ($\text{CLIR@10} = 0.000$, bar not visible) and e5-large-instruct (0.077) from the seven non-degenerate models (lowest SapBERT 0.179). Source: chem-patents extra_two_tax_degeneracy.

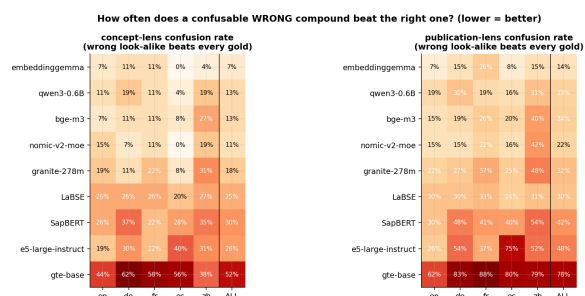


Figure 12: Confusion rate (a look-alike beats all gold), model $\times$ language, both lenses; 14–78% on the publication lens. Source: alias-graph key findings, Fig. 2.

key_findings/EXECUTIVE_SUMMARY.md files
this paper cites.

Generation-pipeline validation (C5). A human annotator reviewed a sample of generated queries; the quantitative human-rating summary and the LLM auto-grader’s strictness relative to humans are summarized in the system description and will be folded in here once dumped to reports/.

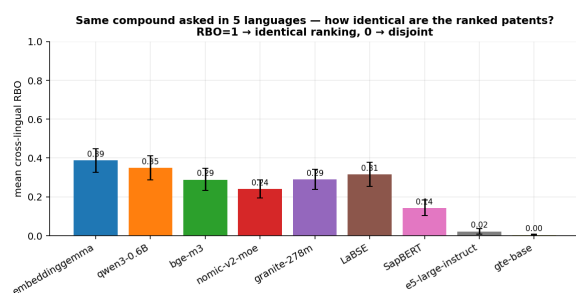


Figure 13: Same compound, five languages: cross-lingual RBO. The ceiling is 0.39 (embeddinggemma). Source: alias-graph key findings, Fig. 1.

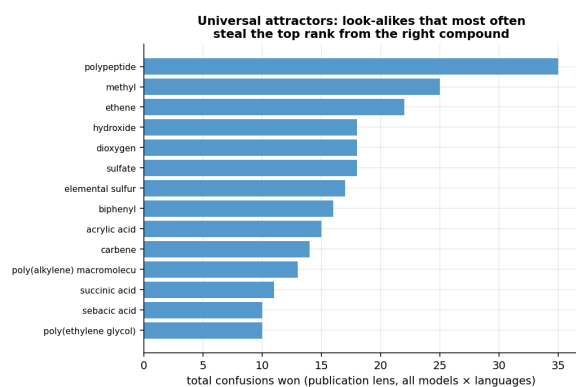


Figure 14: Universal attractors: look-alikes that most often steal the top rank. Source: alias-graph key findings, Fig. 5.

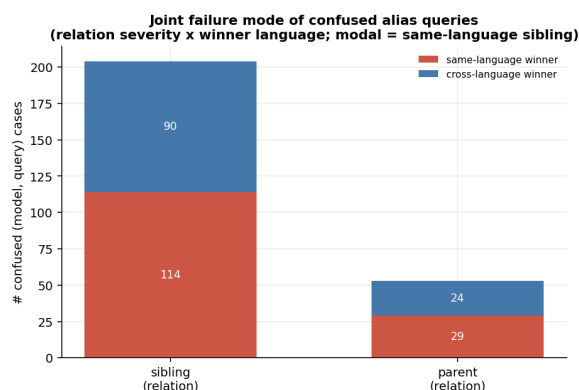


Figure 15: Joint failure modes among confused cases (alias-graph): the modal failure is a same-language sibling (44.4%). Source: alias-graph extra_joint_failure.

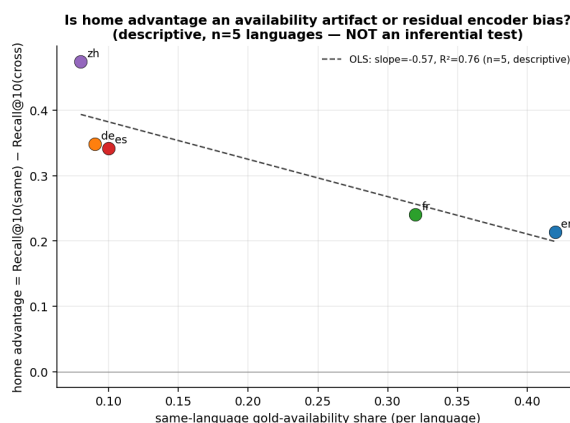


Figure 16: Home advantage vs. gold-availability share per language (alias-graph). The slope is -0.57 ($n=5$, descriptive): home advantage does *not* track availability, so it is residual encoder bias. Source: alias-graph extra_availability_residual.

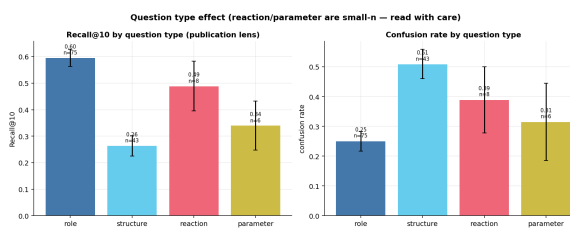


Figure 17: Structure questions are the trap ($R@10$ 0.26, confusion 51%) vs. role questions (0.60, 25%). Source: alias-graph key findings, Fig. 6.

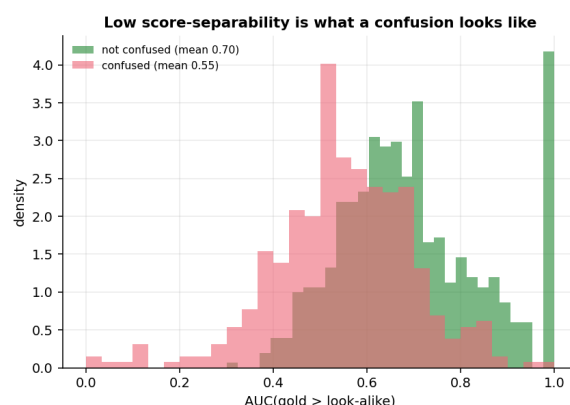


Figure 18: Confusion is a separability collapse: AUC 0.55 confused vs. 0.70 otherwise. Source: alias-graph key findings, Fig. 8.

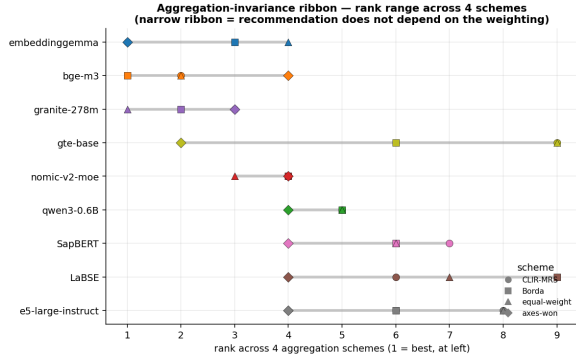


Figure 19: Model rank under four aggregation schemes. embeddinggemma’s rank ranges [1, 4]; the ordering is aggregation-sensitive. Source: chem-patents extra_aggregation_invariance.

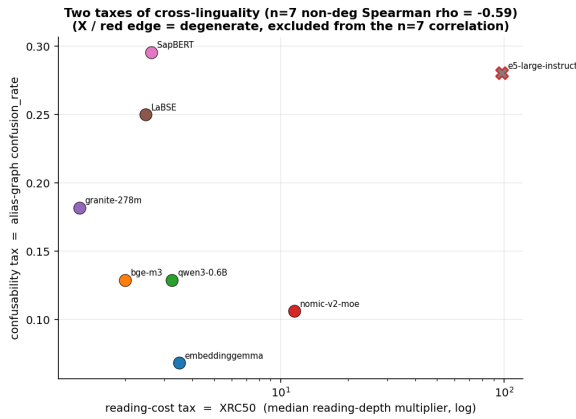


Figure 20: The two line-items of the cross-lingual tax per model: the reading-cost tax (XRC50, cross-lingual benchmark) against the confusability tax (confusion rate, alias-graph benchmark). Across the seven non-degenerate models the two are only weakly and inversely rank-correlated ($\rho = -0.59$, $n=7$, $p=0.16$, n.s.), so neither is a proxy for the other. Source: chem-patents extra_two_tax_degeneracy.

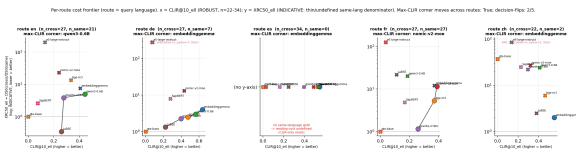


Figure 21: Per-route cost frontier: the $(XRC50_\ell, CLIR@10_\ell)$ frontier recomputed per query language. The capability corner moves—three distinct max-CLIR corners across five routes (en→qwen3-0.6B, de/es/zh→embeddinggemma, fr→nomic-v2-moe)—and the recall-only pick flips on 2/5 routes (en, fr). The $CLIR@10_\ell$ axis is robust; the per-route XRC axis is *indicative* (es undefined, $n_{\text{same}}=0$, not imputed). embeddinggemma stays the single-model default (wins 3/5 routes incl. the hardest); a per-route router is upside headroom. Source: chem-patents extra_per_route_frontier.

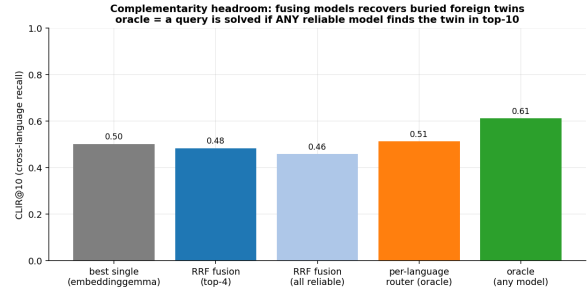


Figure 22: Best-single vs. fusion vs. router vs. oracle (cross-lingual); oracle CLIR@10 0.61, but untuned RRF does not beat the best single model. Source: chem-patents key findings, Fig. 12.

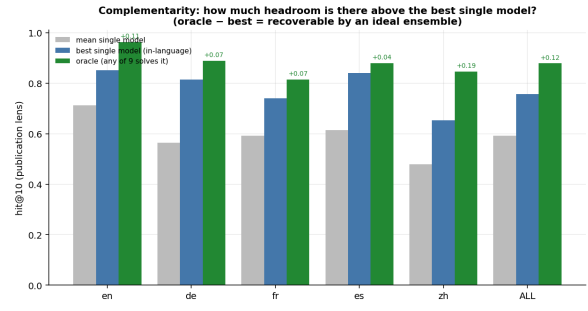


Figure 23: Oracle vs. best single model, per language (alias-graph); 88% vs. 76%, Chinese has the largest headroom. Source: alias-graph key findings, Fig. 7.

scalar	point	resample / stability (n)
separability r	0.96	$P(r>0)=0.9997$; CI [0.73, 1.00] ($n=7$)
XRC50 egemma	3.5	CI [0.91, 12.0] (finite, wide)
XRC50 bge-m3	2.0	CI [0.53, 7.0] (finite, wide)
XRC50 granite	1.25	CI [0.28, 12.25] (finite, wide)
ARI@100 gap (q-e)	0.004	CI [-0.174, 0.176]; $P=0.519$
partial $r \mid R@10$	0.29	$p=0.57$ (n.s.); zero-order 0.96

Table 3: Resample robustness of the body’s small- n scalars. Separability r is reported by its sign (wide CI at $n=7$); the three frontier XRC50 CIs are finite but wide (censored-draw fraction = 0); the ARI@100 gap (qwen3-0.6B—embeddinggemma) CI straddles zero (a tie); and the separability–CLIR partial r controlling Recall@10 is non-significant. Source: chem-patents extra_robustness_appendix.